

How do neutrino event generators work

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- How does an event generator actually “simulate” anything
- The answer is “Monte Carlo” (often called “MC”)
- The Monte Carlo method is simple and effective (but it can be inefficient in some cases)

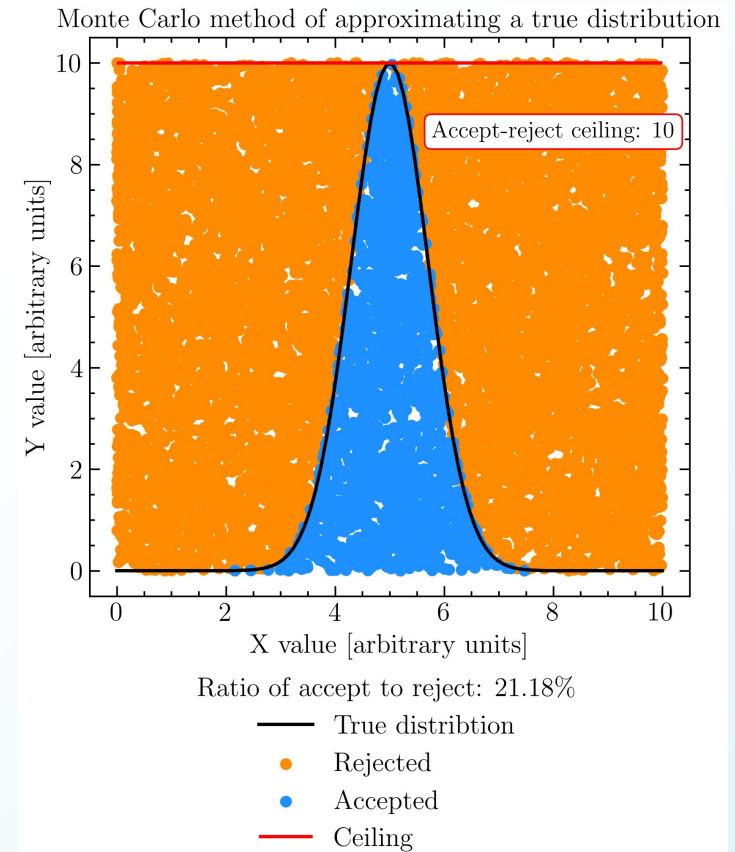
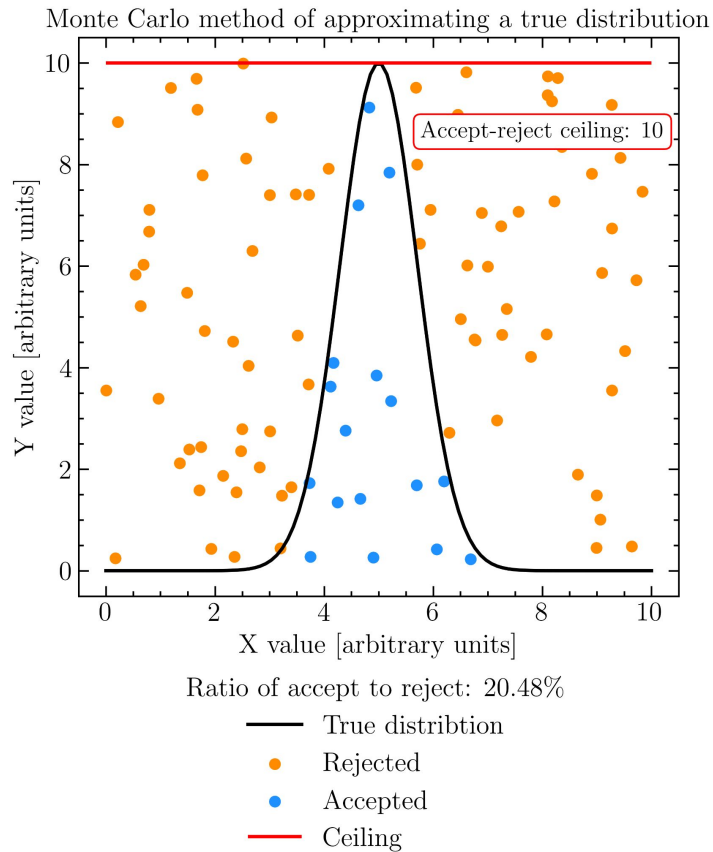
1. Pick a random number between a min and max
 - a. $\text{Rand} = [\text{min}, \text{max}]$
2. Uniformly sample particle kinematics (such as momentum, direction)
3. Calculate the differential cross section with these kinematics (xsec_calc)
4. Check $\text{Rand} < \text{xsec_calc}$
 - a. If true: the event is "accepted" and the information is saved
 - b. If false: the event is "rejected" and begin from step 1 again

- **This is often called "rejection sampling"**

Monte Carlo method

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- The choice of min and max can drastically impact your result!

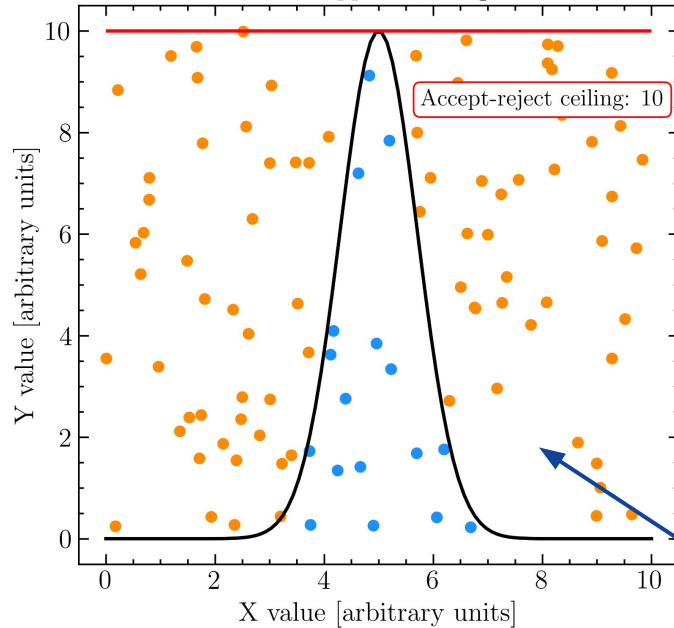


Monte Carlo method

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- The choice of min and max can drastically impact your result!

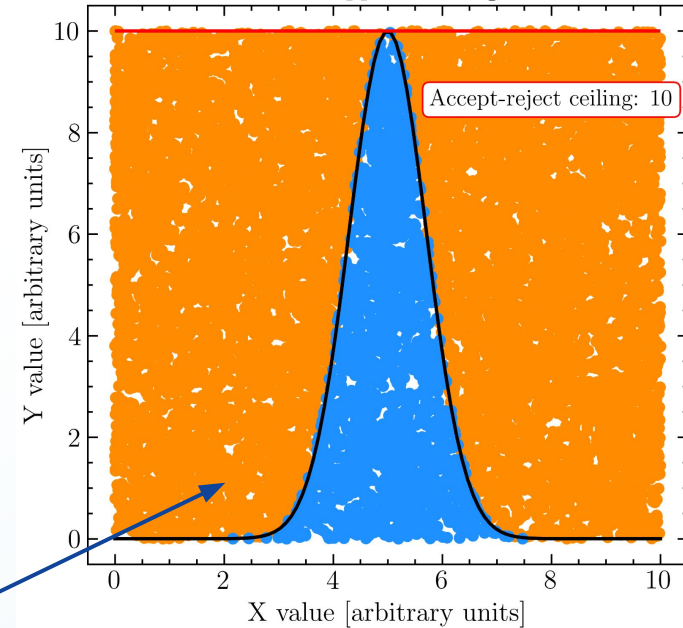
Monte Carlo method of approximating a true distribution



Ratio of accepted points: 22.48%

— True distribution
● Rejected points
● Accepted points
— Ceiling

Monte Carlo method of approximating a true distribution



Ratio of accepted points: 21.18%

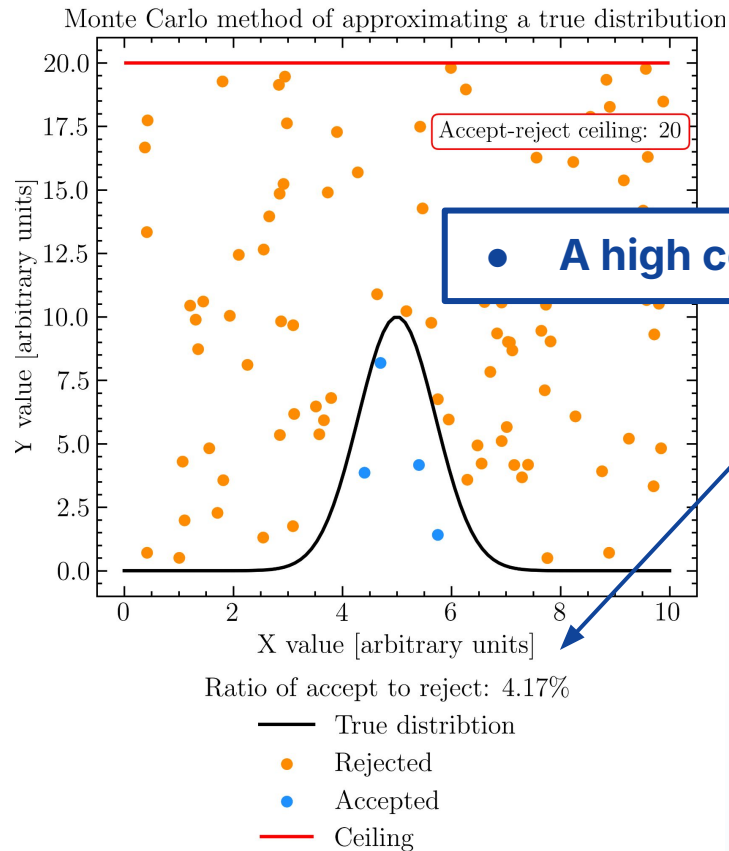
— True distribution
● Rejected points
● Accepted points
— Ceiling

- Low number of events means that you do not accurately replicate the distribution**

Monte Carlo method: too high max

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- If the max is too high, your efficiency will be small!



- A high ceiling will reduce efficiency

